

Secure Exit

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ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT2-LEVEL EXAM 2

Level Exam

</> Java

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QUESTIONS

LEVEL 2 — MEDIUM

Q1

Array



Q2

Numbers



Q3

Control



Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Nth Max — Array

✓ Submitted

Write a Program to Find the Nth Largest Number in a array.

Sample Input:

List : {14, 67, 48, 23, 5, 62}

N = 4

Sample Output:

4th Largest number: 23

Testcases:

1. N = 0
2. N = -5
3. N = 1
4. N = M
5. N = %

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in)
6         try{
7             int n=sc.nextInt();
8             if(n<0)
9                 throw new IllegalArgumentException
10             int[] arr=new int[n];
11             for(int i=0;i<n;i++){
```

```
11         arr[i]=sc.nextInt();
12     }
13     Arrays.sort(arr);
14     int k=sc.nextInt();
15     if(k<=0)
16         throw new IllegalArgumentException
17     int max=arr[n-k];
18     System.out.println(k + "th" +
19 }
20 catch(Exception e){
21     System.out.println("Invalid i
22 }
23 }
24 }
```



Output

Invalid input. Enter valid numbers.



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Q1

Array



Q2

Numbers



Q3

Control



Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Prime and Composite — Numbers

Submitted

Write a program to count all the prime and composite numbers entered by the user.

Sample Input:

Enter the numbers

4

54

29

71

7

59

98

23

Sample Output:

Composite number:3

Prime number:5

Testcases:

1. 33, 41, 52, 61,73,90
2. TEN, FIFTY, SIXTY-ONE, SEVENTY-SEVEN, NINE
3. 45, 87, 09, 5.0 ,2.3, 0.4
4. -54, -76, -97, -23, -33, -98
5. 45, 73, 00, 50, 67, 44

Your Code

```
1 import java.util.Scanner;  
2 class main{  
3     public static void main(String[] args)  
4         Scanner sc=new Scanner(System.in)
```

```
5      int n=sc.nextInt();
6      int prime=0;
7      int composite=0;
8      for(int i=0;i<n;i++){
9          int count=0;
10         int num=sc.nextInt();
11         for(int j=1;j<=num;j++){
12             if(num%j==0){
13                 count++;
14             }
15         }
16         if(count==2){
17             prime++;
18         }
19         else if(count>2){
20             composite++;
21         }
22     }
23     System.out.println("Prime numbers");
24     System.out.println("Composite numbers");
25 }
26 }
```



Output

```
Prime numbers=5
Composite numbers=3
```

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QUESTIONS

LEVEL 2 — MEDIUM

Q1

Array



Q2

Numbers



Q3

Control



Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Tax — Control

✓ Submitted

Write a program to calculate tax given the following conditions:

- If income is less than or equal to 1,50,000 then no tax
- If taxable income is 1,50,001 – 3,00,000 the charge 10% tax
- If taxable income is 3,00,001 – 5,00,000 the charge 20% tax
- If taxable income is above 5,00,001 then charge 30% tax

Sample Input:

Enter the income:200000

Sample Output:

Tax= 20000

Testcases:

Test cases:

- 400700
- 2789239
- 150000
- 00000
- 125486

Your Code

```
1 import java.util.Scanner;
2 class main{
3     public static void main(String[] args)
4         Scanner sc=new Scanner(System.in)
```

```
5      int income=sc.nextInt();
6      if(income<=150000){
7          income=150000;
8      }
9      else if(income>150001 && income<=
10         income=income*10/100;
11     }
12     else if(income>300001 && income<=
13         income=income*20/100;
14     }
15     else if(income>500001){
16         income=income*30/100;
17     }
18     System.out.println("Tax="+income)
19 }
20 }
```

Output

Tax=80140

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QUESTIONS

LEVEL 2 — MEDIUM

Q1

Array



Q2

Numbers



Q3

Control



Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Count Chars — Strings Submitted

Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:

Enter * to exit...

Enter any character: W

Enter any character: d

Enter any character: A

Enter any character: G

Enter any character: g

Enter any character: H

Enter any character: *

Sample Output:

Total count of lower case:2

Total count of upper case:4

Total count of numbers =0

Testcases:

1. 1,7,6,9,5
2. S, Q, l, K,7, j, M
3. M, j, L, &, @, G
4. D, K, I, 6, L, *
5. *, K, A, e, 1, 8, %, *

Your Code

```
1 import java.util.Scanner;
2 class main{
3     public static void main(String[] args)
4         Scanner sc=new Scanner(System.in)
```

```
5      int lower=0, upper=0, number=0;
6      while(sc.hasNext()) {
7          String input=sc.next();
8          for(int i=0; i<input.length();
9              char ch=input.charAt(0);
10             if(ch=='*') {
11                 System.out.println("I
12                 System.out.println("I
13                 System.out.println("I
14                 return;
15             }
16             if(Character.isUpperCase(
17                 upper++;
18             else if(Character.isLower
19                 lower++;
20             else if(Character.isDigit
21                 number++;
22             }
23         }
24     }
25 }
```



Output

```
Total count of lower case=2
Total count of upper case=4
Total count of numbers=0
```



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